



Association between infection prevention and control staffing levels and healthcare-associated infection prevalence: results from the 2022 Belgian European Centre for Disease Prevention and Control point prevalence survey

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SUMMARY

Objectives: The objectives of this study were to describe infection prevention and control (IPC) staffing levels in Belgian hospitals and assess associations with healthcare-associated infection (HAI) prevalence.

Methods: This cross-sectional study analysed data from the Belgian point prevalence survey (PPS) of HAIs and antimicrobial use (Sept–Nov 2022), conducted within the European Centre for Disease Prevention and Control PPS framework. IPC staffing was measured as full-time equivalents (FTEs) for IPC nurses and physicians. Hospital-level characteristics (bed occupancy, intensive care unit presence, and device use) and aggregated patient data were derived from the PPS. Associations between IPC staffing and overall and specific HAI prevalence were examined using multi-variable models adjusted for patient and treatment characteristics.

Results: Median IPC staffing was 1.0 FTE for nurses and 0.3 for physicians (0.83 and 0.30 per 250 beds). Smaller hospitals (<200 beds) reported higher combined IPC staffing per 250 beds than larger hospitals. No association was found between IPC staffing and overall HAI prevalence. However, higher combined staffing was significantly associated with lower prevalence of COVID-19 infection (adjusted odds ratio [ORa]: 0.21; 95% confidence interval [CI]: 0.10–0.42), gastrointestinal infections (0.40; 0.18–0.88), and device-related infections including central line-associated bloodstream infections (0.50; 0.33–0.78) and catheter-associated urinary tract infections (0.60; 0.38–0.95).

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Conclusions: IPC staffing varies widely across Belgian hospitals. Increased staffing is linked to reduced prevalence of certain HAIs. Adequate IPC resources may strengthen infection prevention capacity. Further research should confirm these associations and define optimal staffing strategies.

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Introduction

Healthcare-associated infections (HAIs) are among the most frequent and serious adverse events that occur in healthcare facilities and represent a major global public health issue (Haque *et al.*, 2018). The prevalence of HAIs is associated with prolonged hospital stays, long-term disability, increased antimicrobial resistance, excess mortality, and a substantial financial burden for healthcare systems and patients [1–3]. In Europe, HAIs lead to an estimated 16 million additional hospital days, 37,000 attributable deaths, and 110,000 deaths in which infection is a contributing factor, with direct annual costs of approximately €7 billion [4,5].

According to the most recent European Centre for Disease Prevention and Control (ECDC) point prevalence survey (PPS), 7.1% of hospitalized patients in European countries acquired at least one HAI in 2022–2023, with the national prevalence ranging from 3.1% to 13.8% [6]. This variability suggests that the burden of HAIs is influenced by multiple factors, including patient case mix, indirect antimicrobial use, national infection prevention and control (IPC) policies, and staffing resources dedicated to IPC [7–9]. In Belgian hospitals, HAI prevalence ranged from 1.7% to 24.9% in 2022, a difference unlikely to be explained solely by patient or facility characteristics and potentially linked to variability in IPC staffing levels across institutions [10,11].

Adequate IPC staffing and robust IPC training are associated with better adherence to evidence-based infection prevention practices and lower levels of antimicrobial resistance in hospitals [12–15]. However, IPC teams face increasing pressure due to expanding surveillance and reporting requirements and the need for continuous staff education and epidemic preparedness [16]. The rise of multi-drug-resistant organisms (MDROs) such as carbapenemase-producing Enterobacterales and vancomycin-resistant enterococci, combined with the overuse of antibiotics, further intensifies the IPC team's workload [17–20]. In addition, the COVID-19 pandemic placed IPC professionals under unprecedented stress, highlighting gaps in preparedness and the need for sustainable IPC capacity [21,22].

To support patient and healthcare worker safety, the World Health Organization (WHO) recommends one full-time IPC professional per 250 hospital beds [23]. However, these recommendations are based on the study on the efficacy of nosocomial infection control (SENIC) conducted in the 1970s [24], when healthcare delivery and patient complexity differed substantially [25]. Despite these limitations, IPC staffing levels remain highly heterogeneous across Europe, with no universally accepted standard [26]. Recent ECDC data report a median of 1.25 IPC nurse full-time equivalents (FTEs) and 0.43 IPC doctor FTEs per 250 beds, while 10% and 18% of hospitals, respectively, reported none [6]. However, this descriptive

report did not examine potential associations between IPC staffing levels and HAI prevalence. In Belgium, each hospital receives funding to guarantee at least one FTE of IPC nursing staff and 0.5 FTEs of IPC doctors. The adjustment of this funding per hospital is calculated based on the number of beds and weighted by a coefficient that reflects the type of bed, with the highest coefficients being applied to specialized units such as intensive care units (ICUs), neonatal units, or infectious disease treatment units [27,28].

Despite growing recognition of the importance of IPC resources, significant gaps remain in the literature. Few studies have quantitatively assessed the relationship between IPC staffing levels and HAI outcomes on a national scale. Most existing research has focussed on IPC practice implementation or antimicrobial resistance rather than infection prevalence. Moreover, current staffing recommendations rely on outdated evidence and may not reflect modern healthcare complexities and emerging microbial threats. This lack of contemporary, evidence-based benchmarks limits the ability to guide IPC resource allocation and policy planning. In this context, this study aimed to characterize IPC staffing levels in hospitals and to investigate the associations between IPC staffing levels and HAI prevalence, overall and by type of HAI.

Methods

Study design and setting

This cross-sectional study used data from the Belgian PPS of HAIs and antimicrobial use, coordinated by Sciensano in September–November 2022, as part of the ECDC-PPS 2022–2023 framework [29]. To ensure representativeness of the Belgian hospital population, a systematic sampling design was applied. The sampling framework consisted of 102 administrative hospital groups, obtained from the Federal Public Service Health, Food Chain Safety and Environment Organization [11]. Hospitals were ranked by type, bed capacity, region/province, and ownership. The sampling interval was determined by dividing the total number of acute hospital groups by the number to be sampled ($N=34$). Two substitutions per selected hospital were planned in case of refusal. Incentives of €2 per included patient were provided.

A total of 35 acute hospital groups participated in the 2022 Belgian ECDC PPS, corresponding to a nationwide participation rate of 34.5%. Among these, 30 merged acute hospital groups reported their data at the level of individual hospital sites, resulting in 44 distinct sites. In contrast, 5 multisite hospital groups submitted fully aggregated data for all sites within their organization, representing 12 sites in total. Because aggregated group-level data could not be disaggregated by site, these five submissions were analysed as single hospital-level observations. Consequently, the final analytical dataset

comprised 49 hospitals: 44 site-specific observations and five aggregated multisite hospital groups [11].

The study was reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology guidelines, as endorsed by the Equator network [30].

Participants

All wards within participating hospitals were eligible, including chronic care and long-term care wards, acute psychiatric wards, and neonatal ICUs. Accident and emergency departments were excluded, except for observation wards with patient stays >24 h. Patients present on the ward at or before 8:00 a.m. who had not been discharged by the time of the survey were eligible. Day cases (e.g. same-day surgery, outpatient visits, and dialysis) were excluded. Patient-level data were aggregated at hospital level before the analysis.

Data collection

Data were collected according to the ECDC patient-based PPS protocol for HAIs and antimicrobial use [29]. Sciensano organized a half-day online training session for hospital teams to ensure standardized data collection. Each ward was surveyed on a single day, with the overall hospital survey period limited to 2 to 3 weeks. Hospital-level information was recorded using a standardized form, which captured details on hospital type, size, bed numbers, patient counts, and selected structural and process indicators, including the FTE of IPC nurses and physicians. Reported FTEs for IPC nurses and physicians were verified against national surveillance of IPC quality indicators in acute care hospitals; discrepancies were clarified with hospital staff.

For all eligible patients, data on demographics, risk factors, antimicrobial use, and HAIs were collected.

Variables and data sources

For this study, anonymised hospital-level data were available. Hospital-level variables covered four main domains: structural characteristics, staffing levels, IPC strategy, and patient-related characteristics. Structural characteristics included the number of acute, ICU, neonatal, and total beds, hospital type and ownership, total discharges, patient-days, and bed occupancy rate.

IPC staffing variables included the FTE numbers of IPC nurses and physicians, reporting both as absolute numbers and as FTEs per 250 beds, in accordance with WHO recommendations. The combined IPC staffing level was defined as the sum of the number of nursing and physician IPC staff members per 250 beds. According to Belgian legislation, IPC nurses were defined as nurses with specialized training in infection control or hospital hygiene and IPC doctors as physicians with formal IPC training. IPC strategy indicators included hand hygiene (HH) opportunities and blood cultures per 1000 patients [31].

Aggregated patient-level data per hospital included demographic and clinical characteristics. Demographic variables included the number of patients by age category (0–2, 3–17, 18–35, 36–65, and ≥65 years) and by gender (female, male, or other). Clinical characteristics included the median length of stay, the number of patients who had undergone surgery since admission, and the distribution of patients according to

McCabe score categories (nonfatal, ultimately fatal, rapidly fatal, and unknown). Data on device use included the number of patients with a central venous catheter, urinary catheter, intubation, or vasopressor treatment on the survey day.

HAI data were recorded both overall and by infection type for each hospital. According to the ECDC definition used for overall HAI, an infection was considered active if signs and symptoms were present on the survey day or had occurred previously and the patient was still receiving treatment. The onset of symptoms had to occur on day 3 or later of the current admission (day of admission = day 1) or within 48 h after readmission following discharge or transfer from another healthcare facility [29].

Infection types were recorded according to ECDC definitions and included pneumonia – excluding ventilator-associated pneumonia (VAP), COVID-19 infection, surgical-site infection (SSI), urinary tract infections (UTIs) – excluding catheter-associated urinary tract infection (CAUTI), bloodstream infections (BSIs) – excluding central line-associated bloodstream infection (CLABSI), gastrointestinal infections (*Clostridioides difficile* infection [CDI] and intra-abdominal infection), skin and soft-tissue infections, bone and joint infections, and systemic infections. Device-associated infections were also identified following ECDC criteria and included VAP, CAUTI, and CLABSI [29].

Statistical methods

Qualitative variables are presented as absolute number and relative frequency (N [%]). Quantitative variables are described as medians (Mdn) and interquartile ranges (IQR, p25–p75). Symmetry was assessed using the Kolmogorov–Smirnov test. Comparisons of quantitative variables were made using Mann–Whitney and Wilcoxon tests.

Analyses were performed at the hospital level using aggregated data. IPC staffing levels were studied using absolute FTEs and per 250 beds, separately for nurses and physicians, and combined.

Associations between IPC staffing level (independent variable) and HAI prevalence overall and by type (dependent variables) were assessed using Spearman's rank correlation coefficient (ρ). Strength was interpreted as weak (<0.3), moderate ($0.3–0.5$), or strong (>0.5) [32]. Generalized linear models (GLMs) with a logistic link function were applied. Robust standard errors were used. Backward elimination was used for variable selection, retaining IPC staffing – defined as the combined number of IPC nurses and physicians per 250 beds – in all models. Control variables included mean length of stay, McCabe score distribution, presence of invasive devices, vasopressor use, HH opportunities, and blood cultures performed per 1000 patients. Results are reported as odds ratios (ORs) and adjusted odds ratios (ORAs) with 95% confidence intervals (CIs), where ORa was calculated including all previously identified control variables. P values <0.05 were considered statistically significant. Analyses were performed using STATA v18.0 (Stata: Statistical software for data science; Texas) and R v4.1.2.

Results

Participating hospitals and patient characteristics

The study included 49 hospitals, representing a total of 15,756 beds. The median number of beds per hospital was 257

[IQR: 183–415]. Most hospitals were primary care (57.1%) and privately owned (65.3%). Nearly four out of five had an ICU (79.6%), and almost half had a neonatal unit (46.9%). Median overall occupancy rate was 75% [65–82], with ICU occupancy at 67% [50–80]. The median proportion of single rooms was 38% [25–49].

Among the total 10,142 patients surveyed, 53.9% were aged 65 years or older and a similar proportion were female (53.9%). Most admissions were medical (71.2%). Across hospitals, the median length of stay at the hospital level was 11.8 days [IQR: 10.1–19.6]. According to the McCabe score of underlying conditions, the majority of patients had nonfatal illnesses (69.3%), while 7.2% had rapidly fatal and 18.6% had ultimately fatal conditions. On the day of the survey, 11.8% of patients had a central venous catheter, 14.4% had a urinary catheter, 2.0% had an endotracheal tube, and 0.5% were receiving vasopressor therapy (Table I).

IPC staffing

The median number of FTE IPC nurses per hospital was 1.0 [0.7–1.5] or 0.86 [0.6–1.4] per 250 beds. For IPC physicians, the median was 0.3 [0.2–0.6] FTEs, both in absolute terms and per 250 beds. Combined IPC staffing (nurses and physicians) reached a median of 1.5 [1.1–2.0] FTEs per hospital and 1.2 [1.0–1.8] per 250 beds, with substantial variability across hospitals (Figure 1).

For the combined IPC staffing level (nurses and physicians per 250 beds), no significant differences were observed based on hospital type, ownership status, or the presence of an ICU or neonatal unit. Staffing was, however, associated with hospital size: smaller hospitals (<200 beds) reported higher median staffing (1.6 [1.2–2.1]) than larger hospitals (1.1 [0.9–1.4], $P=0.004$) (Supplementary Table I).

Associations between IPC staffing and HAI

Correlation analyses at the hospital level showed no significant overall association between IPC FTEs reported both as absolute numbers and standardized per 250 beds (nurses, physicians, or combined) and the prevalence of HAIs (Figure 2).

For the combined IPC staffing level (nurses and physicians per 250 beds), when examining specific types of HAIs, higher staffing was associated with lower prevalence of several infection types. Significant associations were observed for COVID-19 (ORa: 0.21, 95% CI: 0.10–0.42), gastrointestinal infections (0.40, 0.18–0.88), and device-related infections including CLABSIs (0.50, 0.33–0.78) and CAUTIs (0.60, 0.38–0.95). No significant associations were observed for pneumonia (0.53, 0.26–1.07), urinary tract infections (0.75, 0.48–1.14), and systemic infections (0.63, 0.31–1.37). No associations were observed for BSIs, SSIs, or VAP (Table II).

Discussion

This study assessed IPC staffing levels in Belgian hospitals and the relationship between staffing levels and HAI prevalence. Several key observations emerged.

First, IPC staffing levels substantially varied across hospitals, particularly for IPC nurses. No significant differences were observed according to hospital characteristics, except that

Table I
Characteristics of included patients and hospitals

Variables	N (%)
Hospital characteristics	
Number of hospitals	49
Number of beds	257 [183–415]
Total beds	15,756
Number of ICUs	39 (79.6)
Number of neonatal units	23 (46.9)
Hospital category	
Primary	28 (57.1)
Secondary	12 (24.5)
Tertiary	3 (6.1)
Specialized	6 (12.2)
Hospital type	
Private	32 (65.3)
Public	17 (34.7)
Single room (%)	38 [25–49]
ICU occupancy rate (%)	67 [50–80]
Occupancy rate (%)	75 [65–82]
Patient characteristics	
Total N patients surveyed	10,142
Age category (years)	
0–2	786 (7.7)
3–17	357 (3.5)
18–35	797 (7.9)
36–65	2739 (27.0)
66 and above	5463 (53.9)
Gender	
Female	5469 (53.9)
Male	4658 (45.9)
Unknown	15 (0.1)
Mean length of stay per hospital (days)	11.8 [10.1–19.6]
Admission category	
Surgery	2865 (28.2)
Medical	7219 (71.2)
No data	58 (0.6)
McCabe score	
Nonfatal	7026 (69.3)
Rapidly fatal	727 (7.2)
Ultimately fatal	1882 (18.6)
Unknown	505 (5.0)
Patient with a central venous catheter	1194 (11.8)
Patient with a urinary catheter	1465 (14.4)
Patient with an endotracheal tube	207 (2.0)
Patient with a vasopressor	52 (0.5)

Data are provided as n (%) or median [interquartile range].
ICU, intensive care unit.

smaller hospitals (fewer than 400 beds) had significantly higher staffing than larger hospitals. This difference may partly reflect the 2007 revision of IPC team funding in Belgium, which allocates resources based on bed-type coefficients (e.g. ICU, neonatal, or surgical) and guarantees minimum staffing requirements of at least 1.0 FTE IPC nurse and 0.5 FTE IPC physicians per hospital. As a result, smaller hospitals may receive comparatively higher support per bed due to these minimum requirements rather than a dedicated small-hospital subsidy [28,31]. However, variability is likely multi-factorial. In addition to funding mechanisms, participation in surveillance

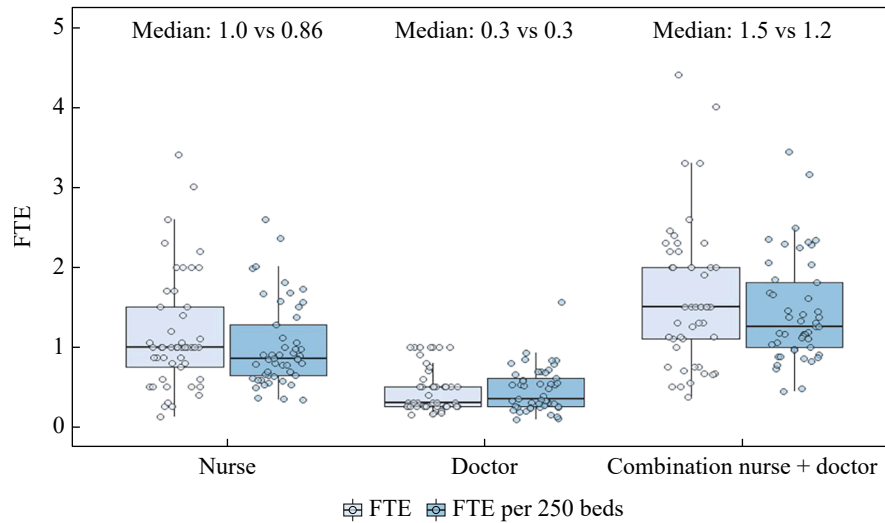


Figure 1. Infection prevention and control staffing levels, overall and by hospital beds
FTE, full-time equivalent.

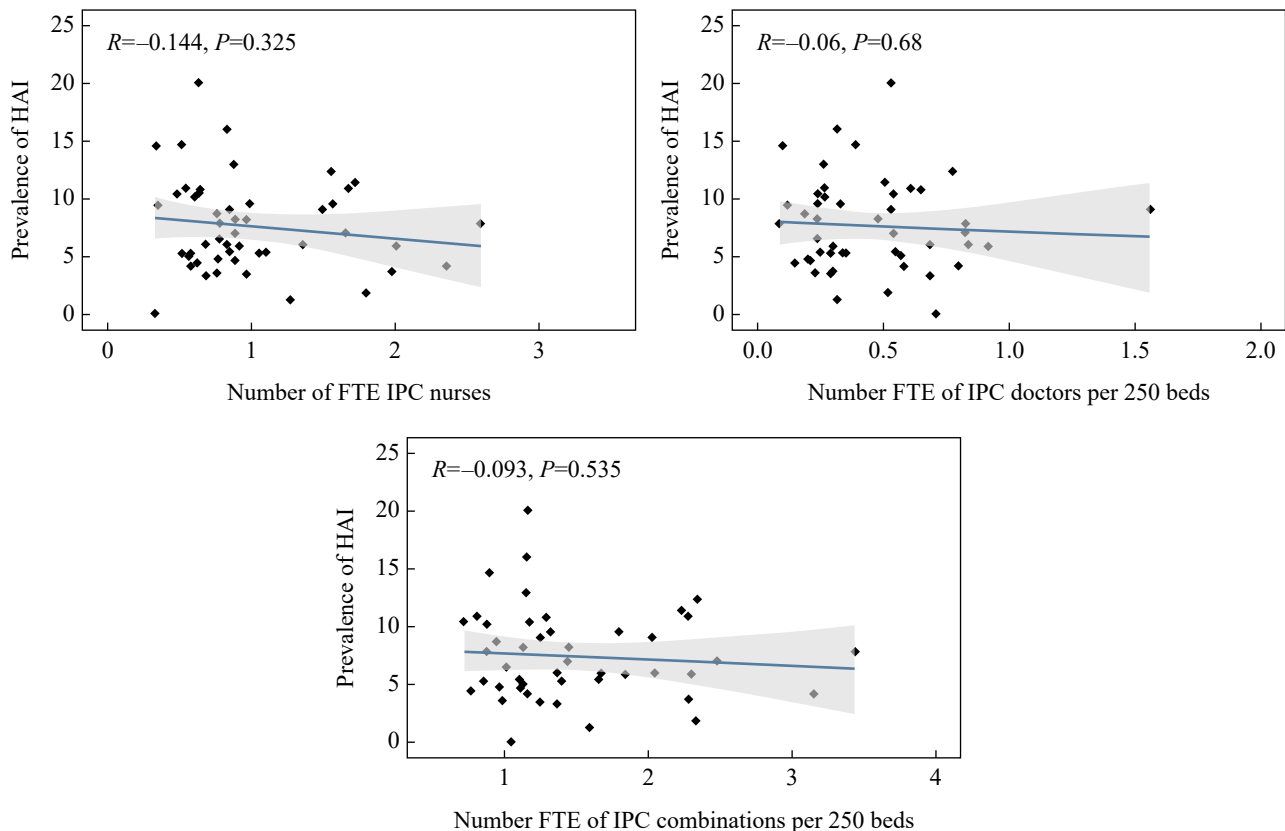


Figure 2. Correlations between infection prevention and control staffing levels and the prevalence of healthcare-associated infections HAI, healthcare-acquired infection; FTE, full-time equivalent; IPC, infection prevention and control.

networks (e.g. SSI, VAP, and CDI) and hospital management priorities may have influenced IPC staffing levels, despite all hospitals in our sample having a validated IPC plan approved by the hospital chief executive officer [28,31]. Such heterogeneity in IPC staffing is not unique to Belgium and has been reported in other countries [33,34]. According to European data, Belgium's IPC staffing recorded in 2024 was above the European median,

though some countries, such as Finland, reported even higher staffing levels [6]. Given differences in IPC staffing levels, assessing workload is essential to ensure both patient safety and the wellbeing of IPC staff [35–37].

Second, no association was found between IPC staffing and overall HAI prevalence, in either univariate or multi-variable analysis. This contrasts with some recent studies suggesting a

Table II

Multi-variable analysis of associations between infection prevention and control staffing levels and healthcare-associated infections overall and by type

Healthcare-associated infections, by type	Unadjusted OR [95% CI]	P-value	Adjusted OR [95% CI]	P value
Healthcare-associated infections	0.96 [0.83–1.09]	0.520	0.96 [0.88–1.04]	0.290
By type				
Pneumonia, non-ventilator-associated pneumonia	0.64 [0.44–0.94]	0.023	0.53 [0.26–1.07]	0.076
Ventilator-associated pneumonia	0.69 [0.25–1.59]	0.337	0.97 [0.45–2.04]	0.935
COVID-19	0.17 [0.08–0.36]	<0.001	0.21 [0.10–0.42]	<0.001
Surgical-site infections	0.63 [0.34–1.17]	0.148	0.87 [0.53–1.43]	0.591
Urinary tract infections, non-catheter-associated urinary tract infection	0.76 [0.56–1.03]	0.077	0.75 [0.48–1.14]	0.204
Catheter-associated urinary tract infection	0.64 [0.45–0.91]	0.014	0.60 [0.38–0.95]	0.032
Bloodstream infections, noncentral line-associated blood stream infections	0.91 [0.59–1.39]	0.667	0.96 [0.58–1.59]	0.880
Central line-associated blood stream infections	0.17 [0.03–0.84]	0.029	0.50 [0.33–0.78]	0.002
Gastrointestinal infections	0.49 [0.25–0.97]	0.041	0.40 [0.18–0.88]	0.023
Skin and soft-tissue infections	0.81 [0.43–1.53]	0.516	0.67 [0.27–1.62]	0.372
Bone and joint infections	0.57 [0.25–1.33]	0.198	0.57 [0.29–1.53]	0.341
Systemic infections	0.55 [0.32–0.95]	0.032	0.63 [0.31–1.37]	0.196

OR, odds ratio; CI, confidence interval.

Control variables (mean length of stay, admission category, McCabe score category, presence of central line catheter, urinary catheter, intubation, vasopressor use, number of hand hygiene opportunities, and blood cultures per 1000 patients) were included in each multi-variate model but are not shown in the table.

Values in bold are statistically significant.

higher IPC staffing level may reduce HAI rates [13,14]. Differences may reflect methodological variations, including how staffing levels and HAIs were measured and the types of hospitals included [36]. Limitations of our study, such as the cross-sectional design and reliance on hospital-reported IPC staffing data, may also have contributed to the lack of an observed association. Furthermore, IPC staffing alone cannot fully capture the effectiveness of infection prevention efforts. Elements such as the competence, specific training, experience and autonomy of IPC teams, the hospital's IPC culture, nursing leadership, adherence to evidence-based protocols, participation in surveillance programmes, implementation of multi-modal strategies, and institutional support are likely to influence HAI outcomes [38–41]. Differences in infection types and transmission dynamics may also obscure simple associations between staffing levels and overall HAI prevalence [42]. These findings underscore the need for more comprehensive analyses that integrate quantitative IPC staffing measures with qualitative aspects of IPC programme implementation [43].

Third, higher IPC staffing levels were associated with a lower prevalence of some types of HAI including COVID-19, gastrointestinal infections, CLABSI, and CAUTI. This association may be explained by stricter compliance with infection prevention protocols, supported by rigorous standard precautions such as HH and systematic screening, as well as the rapid implementation of additional precautions during outbreaks [44,45]. These practices are particularly essential for controlling pathogens with high transmission potential, such as SARS-CoV-2 and those responsible for gastrointestinal infections [13,46,47]. Increased IPC staffing levels also contribute to improved management of invasive devices such as central lines and urinary catheters, ensuring sterile insertion techniques and careful monitoring of duration of use, which reduces the risk of device-related infections [48–50]. Overall, adequate IPC staffing strengthens a hospital's capacity to prevent

infections that are strongly influenced by staff behaviours and adherence to procedures. However, the absence of a uniform effect across all HAIs highlights the multi-factorial nature of infection risk, shaped by pathogen characteristics, transmission routes, patient vulnerability, and institutional practices [51,52].

Overall, these results have important implications for policy and hospital management. Hospitals should monitor and optimize IPC staffing, considering not only staff numbers but also competencies, autonomy, and workload [53]. Support through surveillance networks, institutional prioritization, and resources may enhance IPC effectiveness [54]. Targeted strategies may be required for HAIs most influenced by staffing, such as gastrointestinal infections and device-associated infections. Finally, evidence-based benchmarks for IPC staffing should be updated to reflect modern healthcare complexities, emerging MDROs, and expanded IPC responsibilities.

Limitations and strengths

This study has several limitations. First, its cross-sectional design and reliance on hospital-reported IPC staffing data allow description of staffing patterns and exploration of associations with HAI prevalence but do not support causal inferences. Moreover, hospital-reported figures may not accurately reflect the actual time IPC staff members dedicate to infection control activities or their presence in clinical areas. In addition, endogeneity and reverse causality cannot be excluded. Hospitals with higher HAI rates may have increased IPC staffing, while higher staffing levels may also contribute to lower HAI prevalence, and the study design does not allow distinguishing between these directions of association. Second, IPC staffing estimates were restricted to the number of IPC nurses and IPC physicians, and no information was collected on their professional experience,

specific training, formal qualifications, or competencies, which may influence IPC effectiveness and may be associated with observed HAI outcomes. Third, the study only included Belgian hospitals, limiting generalizability to other health-care systems. Fourth, data were aggregated at the hospital level, preventing patient-level or unit-level analysis and limiting adjustment for confounders. Fifth, data represent a single point in time, whereas staffing and HAI prevalence may vary seasonally or with hospital activity. Sixth, in this single PPS, some outcomes (e.g. CLABSI and VAP) had very few events, with many hospitals recording zero cases. Such sparse data can lead to unstable estimates and small-sample bias when using GLMs with a logistic link, even with robust standard errors. Therefore, effect estimates for these rare outcomes should be interpreted cautiously, given the width of the CIs. Seventh, some hospitals in the dataset represented merged multisite organizations, and aggregated data could not be disaggregated by site. These groups may combine acute, chronic, and specialized care sites, resulting in heterogeneous case-mixes and infection risks within a single analytical unit. This within-hospital heterogeneity may dilute associations between IPC staffing and HAI prevalence and introduces additional noise in the estimates. Furthermore, although single-room availability was collected descriptively, it was not included as a control variable in the models. Its potential role for specific infections such as *C. difficile* or COVID-19 could not be evaluated within this aggregated hospital-level design. Finally, potential confounding factors, such as participation in surveillance networks, institutional support, or hospital policies, were not systematically measured.

Despite these limitations, the study has several strengths. It provides a comprehensive overview of IPC staffing across a national sample, including IPC nurses and physicians, and links staffing data with HAI prevalence. It highlights variability, identifies areas for improvement, and offers an evidence base to inform hospital management and policy regarding IPC resource allocation and workforce planning.

In conclusion, this study demonstrates substantial variability in IPC staffing across Belgian hospitals. While no association was found with overall HAI prevalence, higher IPC staffing was linked to lower prevalence of specific HAIs. Adequate staffing appears to enhance hospitals' capacity to prevent infections, particularly those influenced by staff practices and patient care processes. Further research should confirm these associations, explore causal mechanisms, and determine effective staffing strategies across diverse hospital settings. Policies should consider not only the number of IPC staff but also their competencies, autonomy, and workload, alongside institutional support and participation in surveillance programs to strengthen IPC programs, improve patient safety, and ensure sustainable working conditions for IPC personnel.

CRediT authorship contribution statement

A. Bruyneel: Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **K. Latour:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **M. Pearcy:** Validation, Resources, Methodology. **B. Byl:** Validation. **N. Houdart:** Validation. **A. Simon:** Validation. **L. Catteau:** Writing – review & editing, Writing –

original draft, Resources, Project administration, Methodology, Conceptualization.

Conflict of interest statement

None declared.

Ethics statement

Each participating hospital obtained approval from its local ethics committee prior to participation in the point prevalence survey data collection. All data used for the present study were aggregated at the hospital level and identified only by anonymised codes. Therefore, no additional ethical approval was required. All procedures complied with the General Data Protection Regulation (GDPR), ensuring the security and privacy of patient information throughout the study.

Consent for publication

Not applicable.

Commercial affiliations

None declared.

Declaration of Generative AI and AI-assisted technologies in the writing process

The initial draft of the manuscript was written in French and translated into English using Google Translate. The translation was subsequently reviewed and revised by the authors and by a native English-speaking medical writer, Sandy Field, PhD, with additional corrections and editing to ensure accuracy and clarity. The authors take full responsibility for the final content.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhin.2026.03.005>.

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